Single-Writer Multi-Agent Signal Control with a Read-Only Language Layer: A SUMO Case Study of Six Junctions in Tunis Centre-Ville

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***Abstract*—We present CoFlow-5 Synapse, a hierarchical multi-agent system that controls six signalised junctions of Tunis centre-ville in the SUMO microscopic simulator. The design rests on three rules: each traffic light has exactly one writer, a master agent that arbitrates proposals from five junction agents in a fixed priority order (safety, emergency, network directive, late bus, pedestrian, normal); control degrades from Max-Pressure to actuated to fixed-time when sensor data goes stale; and a language layer explains decisions from a structured decision log but has no code path to the simulator. Every explanation must cite log entries, and an answer whose citations do not exist in the log is turned into an abstention. On a network built from OpenStreetMap with synthetic demand, we compare the system with SUMO's fixed-time and actuated plans over three scenarios and three random seeds. The safety rules cut the number of vehicle greens started while pedestrians were still crossing from 331–405 per run (actuated) to 17–19, and mean pedestrian waiting fell in two of three scenarios. Vehicle waiting time was mixed (-24% to $+20\%$ against actuated), emergency vehicles were slower than with SUMO's plans, and no difference was statistically significant with three seeds. We report these results as preliminary and use recent guidance on stochastic benchmarking in SUMO to identify the weaknesses of our own evaluation protocol, chief among them single-seed parameter tuning and uncorrected multiple comparisons.**

***Index Terms*—traffic signal control, multi-agent systems, SUMO, Max-Pressure, large language models, explainability, reproducibility**

I. Introduction

Most signalised junctions in Tunis centre-ville run fixed-time or locally actuated plans. Each controller sees only its own queues: it cannot see the road beyond the junction, an ambulance two junctions away, or a bus running late. Adaptive and learning-based controllers address part of this, but deploying them raises two practical questions that are as important as average delay. First, *who is allowed to change a light?* When several agents (an emergency agent, a transit agent, a learned policy) can each act on the same signal, their commands can conflict. Second, *why did the light change?* Operators and incident reviewers need an answer that can be checked, not a fluent guess.

Large language models (LLMs) are increasingly coupled to traffic simulators to build scenarios and tune parameters from natural language [1]–[3]. In those systems the LLM acts on the simulation. We take the opposite position for signal control: the language layer may read everything the controllers recorded, but must never reach the simulator or the signal heads.

This paper makes four contributions:

1) A hierarchical multi-agent architecture in which a single master per junction is the only writer of its light, junction agents only propose, and a supermaster coordinates the network through expiring directives (Section III).
2) A read-only language layer that answers questions from a decision log, cites the log entries it relies on, and abstains when citations fail a mechanical check (Section IV).
3) A reproducible testbed of six real junctions in Tunis built from OpenStreetMap, with documented limits on lane evidence and demand (Section V).
4) A preliminary evaluation against SUMO's fixed-time and actuated plans, reported with its negative results, and a critique of our own protocol against recent benchmarking guidance [4] (Sections VI–VII).

All results come from simulation with synthetic demand. SUMO emission outputs are proxies, not air-quality measurements, and simulated emergency trip times are not evidence of lives saved.

II. Related Work

A. Multi-agent signal control in SUMO

Agent-based approaches to traffic and transportation are reviewed by Bazzan and Klügl [5]. Azevedo et al. [6] coupled the JADE agent platform to SUMO through the TraSMAPI middleware, binding one agent to each SUMO traffic light while leaving vehicles as ordinary simulated entities. They contrast two organisations: one agent per light under a *super-coordinator*, which synchronises well but has a single point of failure and concentrated computation, and *distributed* coordination, which synchronises poorly but contains failures. They chose the latter and had each Q-learning agent combine its own reward with the distance-weighted rewards of its neighbours.

Our design is a hybrid of the two organisations (Section III). Two other findings of theirs inform our evaluation: a fixed plan with different timings by time of day beat plain Q-learning on queue waiting time, and waiting time and travel time ranked the controllers differently.

B. Max-Pressure and classical adaptive control

Max-Pressure originates in back-pressure scheduling for packet networks [7] and was introduced to signal control by Varaiya [8], with a stability guarantee under bounded demand. Kouvelas et al. [9] added cyclic phase order, fixed cycles and normalised pressure to limit spill-back. SCOOT [10] and SCATS [11] adapt cycle length, splits and offsets for groups of junctions. Riehl et al. [4] provide open TraCI implementations of fixed-order and flexible Max-Pressure and a SCOOT/SCATS-inspired group controller, and argue that SUMO studies too often rely on single runs and project-specific baselines. They recommend 10–20 replications per configuration, paired comparisons on identical seeds, variance-aware reporting and formal hypothesis tests, and show that parameter settings whose means differ by less than one standard deviation cannot be ranked from a single run [4], [12].

C. Learning-based control

Reinforcement learning (RL) for signal control is commonly studied with multi-agent platforms over SUMO [13]. Our optional RL agent uses Fitted Q-Iteration with tree ensembles [14] and, following [6], a reward that includes neighbouring junctions.

D. Language models and traffic simulation

TrafficGPT [15], ChatSUMO [1] and AgentSUMO [2] connect LLMs to traffic tools so that non-experts can build and run scenarios. Li and Ke [3] split the SUMO workflow into Planner, Builder, Demand, Runner and Analyst agents with an orchestrator that keeps generated artifacts in a persistent store, so that later agents receive validated file references instead of relying on conversational memory. Their analyst feeds KPI feedback back to the planner in the style of Reflexion [16]; one repair round raised task success on harder tasks from 30% to 50%, a second added nothing, and on easy tasks extra reflection sometimes hurt. Their agents also modify the network and signal programmes. Our language layer shares the idea of a single validated source of truth but is deliberately barred from acting.

III. System Design

A. Hierarchy

Table I lists the agents. Each of the six junctions has one master and five junction agents. A supermaster runs eight network agents every 5 s (incident detection, transit monitoring and emergency routing every 1 s) and sends directives to the masters. Agents communicate over a publish/subscribe bus with six message kinds (request, proposal, directive, status, publication, notification); every message carries an expiry time, so a silent agent cannot leave a stale request in force.

TABLE I
Agents of the hierarchy. Only masters write lights, through the executor.

Level	Agents and role
Junction ($\times 6$)	**Master**: arbitrates, sole writer of its light. **Traffic light**: proposes the next phase (Max-Pressure by default; predictive or RL optional). **Pedestrian**, **transit priority**, **emergency priority**: send requests. **Sensor health**: flags stale data.
Network	**Supermaster**: runs network agents, sends directives, never writes a light. Agents: demand forecast, incident detection, rerouting, emergency route, corridor, transit monitor, weather, notification.
Language	**Synapse**: explainer, operator copilot, alert writer, verification bridge. Read-only.

B. Single writer and arbitration

Junction agents never act on SUMO. They send proposals and requests to their master. Each simulated second the master resolves pending messages in a fixed order,

$$\text{safety} \succ \text{emergency} \succ \text{directive} \succ \text{late bus} \succ \text{pedestrian} \succ \text{normal}, \quad (1)$$

and passes at most one phase change to an executor that enforces the safety rules before writing the light: a minimum green of 5 s, a 3 s yellow (longer in rain), a pedestrian clearance hold of up to 30 s computed at a design walking speed of 1.07 m/s, and refusal of a green whose exit lane is more than 85% full. Emergency and transit requests are refused, with a logged reason, when the exit is full, when an earlier emergency vehicle is being served, or when a bus is not late. A static check run in continuous integration verifies that only the executor writes lights, that only masters call it, and that junction agents cannot read scenario events or calendar assumptions.

C. Max-Pressure proposal

Let $\mathcal{L}(\phi)$ be the vehicle links given right of way by phase ϕ, with incoming lane $i(l)$ and outgoing lane $o(l)$; let $\mathcal{C}(\phi)$ be the pedestrian crossings it serves and $\mathcal{E}(\phi)$ the incoming edges it releases. The traffic light agent computes

$$p(\phi) = \sum_{l \in \mathcal{L}(\phi)} \left(h_{i(l)} - h_{o(l)}\right) + \sum_{e \in \mathcal{E}(\phi)} b_e + w_{\text{ped}} \sum_{c \in \mathcal{C}(\phi)} n_c, \quad (2)$$

where h is the number of halting vehicles on a lane, b_e a corridor bias sent by the supermaster for a platoon heading to edge e, n_c the number of pedestrians waiting at crossing c, and $w_{\text{ped}} = 0.5$. The downstream count $h_{o(l)}$ is taken from the neighbouring master's status when it is at most 2 s old, otherwise from local sensors. Once the current green ϕ_0 has lasted the minimum green, the agent proposes $\phi^\star = \arg\max_\phi p(\phi)$ if $p(\phi^\star) > p(\phi_0) + m$, with switching margin $m = 0$. This is the acyclic variant of Max-Pressure [8]. Unlike the flexible controller of [4] it subtracts downstream queues, but it currently has no maximum green time (Section VII).

D. Recovery ladder

The master selects its mode from data freshness (stale after 10 s). With all data fresh it uses Max-Pressure; if downstream data is stale it falls back to actuated control (gap-out after 3 s, max-out at 45 s), which needs incoming lanes only; if incoming data is stale it runs a fixed cycle. Each change of mode is logged with a reason code. The ladder is our answer to the single point of failure that [6] attributes to centralised coordination: if the supermaster or a neighbour goes silent, its directives expire and each junction keeps operating on local data.

E. Learned components behind a rule gate

Learned models supply estimates, not commands: pedestrian over-waiting, ambulance and bus arrival times, bus lateness, demand, travel times, rain slowdown and incident detection. All are scikit-learn [17] gradient-boosted trees trained on data recorded from simulations dated in 2025 and tested on scenarios dated in 2026, so that a model cannot memorise the date of Ramadan. Each model has a hand-written rule as fallback, and a model is used only if it beats its rule on held-out data. The optional RL traffic light agent uses Fitted Q-Iteration [14] with $\gamma = 0.9$, a reward of own waiting plus $0.5\times$ the neighbours' mean queue, and a 60 s starvation guard.

F. Decision log

Every master and supermaster decision is appended to a log entry $(id, t, agent, event, reason, tls, details)$. The reason must be one of a fixed table of codes, each with an official plain-language meaning (for example `EMERGENCY_DENIED_EXIT_FULL`: "the exit lane was full, so green would only move the queue"). Logging an unknown code raises an error, so every decision has a reason.

IV. The Read-Only Language Layer

Synapse answers questions such as "Why did J1 hold green at 08:14?" from a finished or live decision log. The static architecture check forbids its modules from importing the simulation backend, sensors, executor or masters, and the operator copilot's "approve" action is only recorded, never applied.

a) Retrieval: Log entries are selected by junction, vehicle and keyword-to-reason-code matching (for example "ambulance" $\rightarrow$ `EMERGENCY*`, `RECOVER*`). No embeddings are used.

b) Generation: With an API key, a hosted LLM receives only the selected entries and the meanings of their reason codes. It may call read-only tools (log search, live state, experiment results) for at most five rounds. It must cite each factual claim as `[#id]` or reply `ABSTAIN`. Without a key, or if the API fails, a deterministic template builds the answer from the same entries. The control loop never waits for the LLM.

c) Verification: Every cited id is checked against the log. An answer citing an entry that does not exist, or citing nothing, is rejected and returned as an abstention. A separate deterministic bridge answers *verify* and *audit* requests: a claim about a decision is accepted only if it matches the recorded time, action, reason code and agent, and the answer is labelled *verified*, *template* or *abstain* with seven provenance fields. Alerts are rewritten into French and Arabic by an LLM, and a rewrite that changes any number is rejected.

The decision log plays the role that the artifact store plays in [3]: a validated record that the language model reads instead of its own memory. The difference is the direction of authority. In [3] the LLM's outputs are executed after validation; here no LLM output is ever executed.

V. Experimental Setup

A. Network

The network covers Tunis centre-ville, built with `netconvert` from an OpenStreetMap extract dated 15 October 2026 (1,335 edges). Six junctions, J1–J6, were selected by a deterministic rule over the OSM graph (degree $\geq$ 3, then signal tags, roundabout, named, degree, major road, node id); they lie on Av. Habib Bourguiba, Av. Habib Thameur, Rue de Rome and neighbouring streets. Signals were set at the six nodes, and other junctions yield. Lane evidence is weak: only 3 of the 21 roads incident to the six junctions carry an OSM lane-count tag, so the other 18 use SUMO defaults. Roads without a speed tag use Tunisia's 50 km/h urban limit. Bus geometry for 13 lines and 16 stops comes from community-maintained data attributed to the operator TRANSTU, with assumed headways.

B. Demand and scenarios

No open traffic counts exist for these junctions, so demand is synthetic: trips generated with SUMO's `randomTrips` from a calendar model at a base of 2,400 vehicles/h plus pedestrians and buses. Twelve scenarios (peaks, Ramadan, rain, incidents, emergencies, sensor faults) are defined; scenario events are hidden from the agents. This paper reports three of them: *morning peak*, *emergency* (ambulances crossing the network) and *incident* (lane blockages). Each scenario was run with seeds 1–3 at a 1 s step in SUMO 1.27.

C. Controllers and metrics

We compare four controllers on identical demand files per seed: SUMO's *fixed*-time plans, SUMO's *actuated* plans, *traffix rules* (the full hierarchy with rule fallbacks only) and *traffix models* (the full hierarchy with trained models and Max-Pressure lights). Metrics are mean vehicle waiting time, mean pedestrian waiting time, emergency vehicle trip time, and the number of vehicle greens started while pedestrians were still on a crossing. Differences are tested with paired t-tests over seeds.

VI. Results

Table II gives the results.

TABLE II
CONTROLLER RESULTS ON THE TUNIS NETWORK (MEAN OF SEEDS 1–3, LOWER IS BETTER, BEST IN BOLD). PERCENTAGES COMPARE TRAFFIX MODELS WITH SUMO ACTUATED. NO DIFFERENCE IS SIGNIFICANT AT $p < 0.05$.

Scenario	Measure	SUMO fixed	SUMO actuated	traffix rules	traffix models
Morning peak	Mean vehicle wait (s)	122.6	118.5	**101.8**	105.9 (-11%)
Emergency	Mean vehicle wait (s)	**62.2**	65.9	68.7	78.8 ($+20\%$)
Incident	Mean vehicle wait (s)	**69.1**	96.1	98.1	73.5 (-24%)
Morning peak	Mean pedestrian wait (s)	140.2	144.1	128.1	**116.3** (-19%)
Emergency	Mean pedestrian wait (s)	139.1	152.2	107.6	**98.3** (-35%)
Incident	Mean pedestrian wait (s)	123.9	**111.1**	118.9	123.3 ($+11\%$)
Emergency	Emergency vehicle trip (s)	**197.0**	213.2	232.3	240.0 ($+13\%$)
All three	Greens started during a crossing (per run)	239–281	331–405	16–19	**17–19**

a) Pedestrian safety: The clearest effect is on greens started while pedestrians were still crossing: 331–405 per run with actuated plans and 239–281 with fixed plans, against 16–19 with either traffix variant, a reduction of about 95%. Because the rule-only and model variants are equal here, the gain comes from the executor's clearance rule and not from learning. It is not zero, so the pedestrian-safety objective is only partly met.

b) Waiting times: Pedestrian waiting fell against actuated in the morning peak (-19%) and emergency (-35%) scenarios and rose in the incident scenario ($+11\%$). Vehicle waiting was mixed: lower in the morning peak (-11%) and incident (-24%) scenarios, higher in the emergency scenario ($+20\%$). In two of three scenarios the simplest baseline, fixed time, gave the lowest vehicle wait. With three seeds none of the paired t-tests between traffix and actuated, or between models and rules, was significant at $p < 0.05$; the closest was pedestrian waiting in the morning peak, models against rules, -9% ($p = 0.06$).

c) Emergency vehicles: Emergency trips were slower with traffix (240.0 s) than with actuated (213.2 s) or fixed plans (197.0 s), even though the emergency agent requests green ahead of the vehicle. We have not yet identified the cause; candidates are the preparation horizon, the exit-full refusal rule and the extra vehicle queues in this scenario. The emergency objective is not met.

d) Learned components: Seven of eleven models beat their rule on held-out data, for example ambulance arrival time (mean error 43.7 s against 92.6 s for distance over speed) and demand forecasting during Ramadan (0.026 against 0.103). The arrival model only matched its rule, the pedestrian classifier is weak (F1 0.44), and the trained sensor-health classifier lost to its rule (F1 0.75 against 0.98) and was removed. Random search over the Max-Pressure parameters (8 trials) found nothing better than the defaults. After three rounds of Fitted Q-Iteration, the RL light waited 231.4 s (vehicle plus pedestrian) in the final test against 221.9 s for Max-Pressure, about 4% worse, so Max-Pressure remains the default.

e) Language layer: Every logged decision carries a reason code by construction, and in our tests a false claim submitted to the audit endpoint is answered with an abstention. We have not yet measured the explainer quantitatively (Section VII).

VII. DISCUSSION AND THREATS TO VALIDITY

We now examine our own protocol against the guidance of [4], and our design against [6] and [3].

A. Evaluation protocol

Single-seed tuning. The Max-Pressure parameters were tuned by scoring each candidate on one seed. Riehl et al. show that neighbouring parameter values can differ by less than their run-to-run standard deviation (12.48 against 13.04 with standard deviations of about 2.3 over 20 runs, $p = 0.46$), so a single run can rank them either way. Our conclusion that "no candidate beat the defaults" may therefore reflect noise. Candidates should be scored by their mean over a fixed, documented set of seeds.

Too few replications. Three seeds leave a paired t-test with little power, so our "not significant" results are weak evidence of no effect. The recommended range is 10–20 replications per configuration.

Multiple comparisons. Our analysis script tests five controller pairs on three metrics in each scenario, up to 180 tests over all twelve scenarios. At $\alpha = 0.05$, about nine would appear significant by chance even if no controller differed. A family-wise correction such as Holm's [18] is needed, and where normality is doubtful the Wilcoxon signed-rank test [19] should replace the t-test.

Simulation settings. We measure from the first second, without a warm-up period, and use a 1 s step where 0.25 s is recommended for urban scenes. Both may bias waiting times.

Baselines. SUMO's default fixed and actuated plans are weak references, and fixed time won two of our vehicle-wait comparisons. Following [6], a fixed plan with time-of-day timings is a stronger and fairer baseline for learned controllers; following [4], an external Max-Pressure and a SCOOT/SCATS-style group controller would show whether our gains, where present, come from the architecture or from a weak baseline.

No maximum green. Our Max-Pressure mode has no maximum green time, whereas the flexible controller of [4] caps

green at 50 s and our own RL agent at 60 s. A busy approach can therefore keep green while a quiet one starves.

Fairness metrics. Following [6], the standard deviation of trip and waiting times should be reported next to the means, since controllers can trade a lower average for more variable service.

B. Architecture

Azevedo et al. [6] present centralised and distributed coordination as alternatives. Our hierarchy combines them: decisions remain local and single-writer, while the supermaster provides the network view through directives that expire. The *junction-only* controller in our experiment suite (masters without a supermaster) measures what the coordinator adds; it was not part of the three-scenario run reported here.

C. Language layer

Li and Ke [3] found that a single repair round recovered many failed LLM runs while further rounds did not. Our explainer abstains at the first failed citation check; one retry with the error fed back is a cheap improvement. They also note that smaller models break output schemas unless the output is validated, which argues for a structured {`answer`, `citations`, `abstained`} output over citation extraction from free text. Finally, their evaluation (task success, tokens and time to first result across three difficulty levels) is a template for the benchmark Synapse lacks: a fixed set of questions over saved logs, scored on citation correctness, correct abstention, cost and latency, for both the LLM and the template engines.

D. External validity

Demand is synthetic, 18 of 21 junction roads have unverified lane counts, bus headways are assumed and calendar effects are our own assumptions. The results describe this simulated network only.

VIII. Conclusion

We built a six-junction multi-agent signal controller on a real street network in which one master writes each light, control degrades predictably when data fails, and a language layer explains decisions from a log it cannot change. The architecture rules are enforced mechanically, and every decision is logged with a reason. The clearest measured benefit is a 95% reduction in greens started during pedestrian crossings, which comes from the safety rules rather than from learning. Effects on waiting times are mixed and not yet significant, and emergency vehicles were slower than with SUMO's own plans. Our next steps follow from the critique above: tune over several seeds, run 10–20 seeds over all twelve scenarios with corrected tests and a warm-up period, add a maximum green, compare against time-of-day and external adaptive baselines, find the cause of the emergency slowdown, and benchmark the language layer.

Data and Code Availability

Code, network, scenario definitions and the scripts that produce every number in this paper are in the project repository: `scripts/run_experiments.py` and `scripts/analyze.py` for Table II, `models/metrics.json` for the model comparisons, and `scripts/check_architecture.py` for the architecture rules.

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